COURSE 2A

C2-07 CLOUD / FABRIC / ENTERPRISE BI

BEGINNER-FIRST TEACHING BOOK - RC1

Cloud boundaries -> Fabric -> Service -> deployment -> refresh -> governance -> monitoring

THIS BOOK IS YOUR TEACHER

Platform-specific references refreshed 2026-09-10. Internal concepts remain sufficient if a link or trial is unavailable.

This book is your teacher

Enterprise BI is learned through roles, release, refresh, security and incidents - not screenshots alone

THE STUDY LOOP

Why -> mental model -> vocabulary -> useful current source/video -> follow one enterprise scenario -> expected result -> why it works -> fresh design task -> validate -> explain-back -> save attempt -> protected review -> changed retry.

TENANT / TRIAL RULE

Core mastery is possible from the book, workbook and synthetic fixtures. A Fabric trial or organizational tenant is OPTIONAL. Microsoft currently offers a time-limited Fabric trial for eligible tenants/users, but eligibility/features can change and no Gate depends on it.

LIVE-RUNTIME RULE

If you do have Power BI/Fabric access, record runtime evidence separately. If a tenant feature is unavailable, mark ENVIRONMENT BLOCKED. Never fake workspace/gateway/deployment evidence from screenshots in this book.

C2-06 HANDOFF

C2-06 already taught warehouse/lakehouse architecture, lineage and quality. C2-07 does not repeat that body. It adds the cloud operating layer: identity, Fabric workload topology, enterprise Power BI, release environments, gateways/refresh, permissions/governance, monitoring and team handoffs.

Contents / Index

Every canonical C2-07 lesson and actual page number

Prerequisite and tool map

What is assumed, what is new and what can be environment-blocked

ASSUMED

Course 1 Power BI basics; C2-04 lifecycle/RLS/reporting; C2-06 warehouse/lakehouse/lineage concepts. New cloud/Fabric/enterprise-operating terms are defined before use.

CORE TOOLS

Teaching Book + C2-07 Enterprise Operations Workbook + synthetic item/workspace/source/job/access fixtures. These are enough for required reasoning and Gate evidence.

OPTIONAL LIVE TOOLS

Power BI Service / Microsoft Fabric tenant or trial, deployment pipelines, gateway, Monitor and OneLake catalog. Availability depends on tenant, license, capacity, region and organizational policy.

PL-300 POSITION

CLD5-CLD8 directly reinforce current Manage and Secure Power BI skills: workspaces/apps/distribution, gateways/refresh, roles/item/model access/RLS and sensitivity labels. Course 2A remains optional; PL-300 is supplementary, not a hidden DA-to-DE prerequisite.

Practice, review and Gate route

Architecture plan -> operational evidence -> incident/change transfer

Read -> Design -> Fixture check -> Failure case -> Explain -> Review -> Changed retry -> Master

LESSON RULE

Use matching P_CLD# workbook sheet after the exact lesson pages. Save first enterprise plan/evidence before protected review. Close the review before changed transfer.

MINI-LAB

NorthStar Enterprise Analytics Deployment: workspaces/items, builder/consumer access, DEV/TEST/PROD, refresh/gateway, governance, monitoring incident and team handoff.

GATES

A Fleet Maintenance, B Education Enrollment, C Energy Metering. Distinct workspace/source/access/refresh/incident fixtures. Any critical unsafe access or unowned failure blocks pass.

CLD1 - Cloud data fundamentals

Page A - why, outcome, prerequisite, mental model and vocabulary

Business need -> Managed service -> Tenant -> Workspace -> Capacity -> Data items

WHY IT MATTERS

Cloud analytics can feel like a wall of product names. Before Fabric, you need a simple model: who owns the service, where your organization lives, where teams work, what provides compute and where analytical items/data are organized.

WHAT YOU LEARN

Explain cloud responsibility at a beginner level; distinguish tenant, workspace, capacity and item; recognize Fabric as SaaS and avoid confusing 'cloud' with 'someone else's laptop'.

PREREQUISITE

C2-06 workload/warehouse concepts help, but no Azure administration experience is assumed.

NEW WORDS BEFORE WE USE THEM

SaaS: Software delivered and operated as a managed service; you use/configure it rather than managing all infrastructure. Tenant: Organization-level boundary for identities, settings and Fabric/Power BI environment. Capacity: Allocated compute resources used by Fabric workloads; separate concept from OneLake storage.

Place one enterprise BI solution in the cloud

1. Business asks for governed Sales analytics used by 200 employees. Start with the organization/tenant—not with a VM.
2. Create a conceptual Analytics workspace for team-owned Fabric/Power BI items. A workspace is a collaboration/security boundary inside the tenant.
3. Place a semantic model/report and, if needed, a lakehouse/warehouse/pipeline inside the workspace.
4. Capacity supplies compute for Fabric workloads; identity and tenant policies control who can use/create items. OneLake is Fabric's unified logical data lake, not the same thing as capacity.
5. Write which responsibilities remain with Microsoft (managed service/infrastructure) versus your organization (access, data, workspace design, definitions, governance, cost/capacity decisions).

EXPECTED RESULT

You can place tenant -> workspace -> items and explain capacity/identity responsibilities without Azure-infrastructure jargon.

WHY IT WORKED

The mental model separates organization, collaboration boundary, compute and data/items instead of treating 'cloud' as one box.

CLD1 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: a retail company needs Finance and Operations analytics with separate ownership but some shared certified data. Sketch tenant/workspace/capacity/item boundaries.

COMMON BEGINNER MISTAKES

1. Tenant = workspace. | 2. Capacity = storage folder. | 3. Cloud removes responsibility for permissions/governance. | 4. Assuming Fabric requires you to provision Azure resource groups for ordinary use.

STUCK? RECOVERY

Draw four labeled boxes only: Organization/Tenant -> Workspace -> Items, with Capacity beside them as compute. Add OneLake as shared storage foundation. Save the genuine attempt before review.

VALIDATE

Your diagram names owners, access boundary, compute boundary and shared-data path; no product term is used without a job.

CLD1 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

Fabric is currently delivered as SaaS, and Fabric tenants automatically include OneLake. Capacity sizing/licensing can change, so this course teaches responsibilities and boundaries before SKU memorization.

TEACH IT BACK

Explain tenant, workspace and capacity to a manager using organization -> team room -> engine.

PROTECTED REVIEW + CHANGED RETRY

Save first diagram. Changed retry uses a consulting company with two client workspaces and a central governed semantic-model team.

EVIDENCE / MASTERY

Cloud responsibility matrix + tenant/workspace/capacity/item diagram + explain-back.

VISUAL/SOURCE ROUTE: Microsoft Fabric overview is required visual/reference. Trial is optional and never required for mastery.

https://learn.microsoft.com/en-us/fabric/get-started/microsoft-fabric-overview

CLD2 - Compute, storage and identity concepts

Page A - why, outcome, prerequisite, mental model and vocabulary

Identity -> Permission -> Compute -> Storage -> Item -> Action

WHY IT MATTERS

Most enterprise incidents happen because someone confuses 'can the engine run?' with 'is this user allowed?' or 'where does the data live?'. Compute, storage and identity are separate control planes.

WHAT YOU LEARN

Distinguish compute, storage and identity; apply least privilege to people/groups/service principals conceptually; explain why a job may have compute but still fail due to data/credential permission.

PREREQUISITE

CLD1 tenant/workspace/capacity model.

NEW WORDS BEFORE WE USE THEM

Compute: Resources that execute queries, refreshes, pipelines or notebooks. Identity: User, group or application/service principal that requests access. Least privilege: Grant only the minimum permissions needed for the intended job, for the required scope/time.

CLD2 - Follow with me once

Page B - Why a refresh can fail even when capacity is healthy

Why a refresh can fail even when capacity is healthy

1. Capacity is available: the service has compute to run a semantic-model refresh.
2. Refresh identity/connection still needs permission and valid credentials to reach its data source.
3. The workspace role controls what a person/service principal can do with Fabric/Power BI items, but source-system permission can be separate.
4. OneLake/item permissions and semantic-model/RLS permissions can further constrain data access.
5. Create a matrix: Identity -> Resource -> Needed action -> Minimum permission -> Owner. Do not solve access by granting Admin everywhere.

EXPECTED RESULT

A clear diagnosis separating compute availability, item permission, source credentials and data-level security.

WHY IT WORKED

Each failure is tested at the control plane where it can actually occur.

CLD2 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: service principal publishes items, analyst builds reports, viewer consumes app, gateway identity reaches SQL Server. Design minimum conceptual permissions.

COMMON BEGINNER MISTAKES

1. Admin workspace role fixes every source-access problem. | 2. Healthy capacity proves data source is reachable. | 3. RLS replaces all workspace/item permissions. | 4. Use personal credentials for production because they are easiest.

STUCK? RECOVERY

For any 'access failed' issue ask four questions: Who is the identity? Which resource? Which action? Which permission/credential controls that action? Save the genuine attempt before review.

VALIDATE

No identity receives unexplained broad Admin access; source, workspace and data-level permissions are separate.

CLD2 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

Enterprise automation increasingly uses groups and service principals rather than personal accounts. Exact tenant/API policies vary; the durable practice is explicit identity/action/scope/owner.

TEACH IT BACK

Explain why capacity, permission and data credentials are three different reasons a cloud job can succeed or fail.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry removes one employee and requires group-based access without editing every item manually.

EVIDENCE / MASTERY

Identity-action-permission matrix + failure diagnosis + explain-back.

VISUAL/SOURCE ROUTE: Book-led; current workspace-role and Fabric overview references are enough.

https://learn.microsoft.com/en-us/fabric/fundamentals/roles-workspaces

CLD3 - Microsoft Fabric ecosystem

Page A - why, outcome, prerequisite, mental model and vocabulary

Sources -> Data Factory -> Engineering -> Warehouse -> RTI -> Power BI

WHY IT MATTERS

Fabric combines many analytics experiences. Memorizing icons does not help you design a solution; you need to know which workload solves ingestion, engineering, warehouse, real-time or BI consumption problems and how they share OneLake.

WHAT YOU LEARN

Map major Fabric workloads to responsibilities; describe Fabric as end-to-end SaaS analytics; choose the smallest set of workloads needed for a business flow.

PREREQUISITE

CLD1-2 and C2-06 architecture basics.

NEW WORDS BEFORE WE USE THEM

Workload: Fabric experience optimized for a class of analytics tasks. Data Factory: Fabric workload/capabilities for data movement and orchestration. Real-Time Intelligence: Fabric workload for ingesting/analyzing event and streaming data.

CLD3 - Follow with me once

Page B - Map one Sales analytics flow

Map one Sales analytics flow

1. Data arrives from CRM/order sources. Data Factory may orchestrate copy/ingestion.
2. Data Engineering may use Spark/lakehouse transformations when that tool fits the workload.
3. Data Warehouse may serve governed relational dimensional tables queried with T-SQL.
4. Real-Time Intelligence is appropriate when operational event streams require real-time analysis; do not add it to a daily batch just because it exists.
5. Power BI provides semantic/reporting consumption. All required workloads can coexist in Fabric and access shared OneLake data/items according to permissions.

EXPECTED RESULT

A workload map where every Fabric component has a reason and unnecessary services are omitted.

WHY IT WORKED

You design from task -> workload instead of icon -> task.

CLD3 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: IoT maintenance events + daily ERP finance + executive BI. Choose Fabric workloads for each path and explain integration.

COMMON BEGINNER MISTAKES

1. Every solution needs every Fabric workload. | 2. Data Factory = the warehouse. | 3. Power BI Service and Fabric are unrelated platforms. | 4. Real-Time Intelligence for monthly CSV reporting.

STUCK? RECOVERY

Write the verbs: ingest, transform, store/query, stream, model/report. Map one tool only when a verb needs it. Save the genuine attempt before review.

VALIDATE

Every chosen workload solves a named problem; no duplicate/unnecessary workload is added.

CLD3 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

Microsoft currently lists Data Engineering, Data Factory, Data Science, Real-Time Intelligence, Data Warehouse, Databases and Power BI among Fabric experiences. The list can evolve; workload-to-job reasoning is the durable skill.

TEACH IT BACK

Explain Fabric to a beginner as one managed analytics platform containing role-specific work areas over shared data.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry removes streaming need and simplifies the design.

EVIDENCE / MASTERY

Workload decision map + simplified alternative + explain-back.

VISUAL/SOURCE ROUTE: Required: current What is Microsoft Fabric overview; optional Power BI Meets Fabric video.

https://learn.microsoft.com/en-us/fabric/get-started/microsoft-fabric-overview

CLD4 - OneLake, lakehouse and warehouse concepts

Page A - why, outcome, prerequisite, mental model and vocabulary

Tenant OneLake -> Workspace -> Shortcut -> Lakehouse -> Warehouse -> BI/AI

WHY IT MATTERS

C2-06 taught when a warehouse/lakehouse pattern fits. Here the new question is how Fabric organizes and shares those items over OneLake and when shortcuts/shared storage can reduce unnecessary data copies.

WHAT YOU LEARN

Explain OneLake tenant-wide logical storage, workspaces/items and shortcuts; place lakehouse/warehouse items in an enterprise Fabric topology; avoid duplicating C2-06 dimensional-design theory.

PREREQUISITE

C2-06 DWF8-DWF9 plus CLD1-3.

NEW WORDS BEFORE WE USE THEM

OneLake: Fabric's unified logical data lake for the organization. Shortcut: Reference that provides access to data in another supported location without traditional copy/migration. Item: Fabric object such as lakehouse, warehouse, semantic model or pipeline inside a workspace.

One data estate, two teams

1. Central Data Platform workspace owns curated customer/sales data in a Warehouse or Lakehouse depending on workload.
2. Finance BI workspace consumes governed semantic/reporting assets rather than receiving uncontrolled CSV copies.
3. Where appropriate, OneLake shortcuts can expose existing data without duplicating it into another lakehouse; still apply ownership/security.
4. Document which workspace owns the authoritative item and which team only consumes it.
5. Do not use OneLake to erase governance boundaries: shared storage does not mean every user sees every item.

EXPECTED RESULT

A Fabric topology with clear data ownership, shared use and minimized unnecessary copies.

WHY IT WORKED

OneLake unifies storage access while workspaces/items/security preserve organizational control.

CLD4 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: Marketing needs governed Customer 360 from central lakehouse while Finance uses a warehouse and shared certified semantic model. Draw item/workspace/data paths.

COMMON BEGINNER MISTAKES

1. OneLake = one giant folder everyone can read. | 2. Shortcut automatically grants access to every consumer. | 3. Duplicate every dataset per workspace 'for safety'. | 4. Re-teaching warehouse/lakehouse choice without Fabric topology.

STUCK? RECOVERY

Start with authoritative owner and consumer. Ask whether data must be copied, referenced or surfaced as a semantic model. Save the genuine attempt before review.

VALIDATE

For each copy/reference, state why it exists; ownership and permissions remain explicit.

CLD4 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

OneLake currently supports unified tenant storage and zero-copy/shortcut patterns to supported external sources. Exact shortcut/security capabilities can evolve, so design must be validated against current docs before production.

TEACH IT BACK

Explain OneLake as one organizational data-lake foundation, not one unrestricted spreadsheet folder.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry adds an external S3/ADLS source and asks whether a shortcut is preferable to copy.

EVIDENCE / MASTERY

Workspace/item topology + ownership/copy decision + explain-back.

VISUAL/SOURCE ROUTE: Required visual: OneLake overview; optional 'No Walls, Just Data' session.

https://learn.microsoft.com/en-us/fabric/onelake/onelake-overview

CLD5 - Power BI Service in enterprise environments

Page A - why, outcome, prerequisite, mental model and vocabulary

Builders -> Workspace -> Semantic model -> Reports -> App/share -> Consumers

WHY IT MATTERS

Publishing a PBIX is not the same as enterprise delivery. Teams need clear builder and consumer routes, trusted semantic models, distribution methods, ownership and support.

WHAT YOU LEARN

Design workspace collaboration, semantic-model/report ownership, app/direct share choices, content endorsement and consumer access; separate build surface from polished distribution.

PREREQUISITE

C2-04 DAX15 lifecycle basics and CLD1-4.

NEW WORDS BEFORE WE USE THEM

App: Packaged Power BI distribution experience published from a workspace to audiences. Semantic model: Governed analytical model consumed by reports/Excel/other clients. Endorsement: Power BI/Fabric trust signal such as promoted/certified content under organizational policy.

CLD5 - Follow with me once

Page B - From BI team workspace to 500 consumers

From BI team workspace to 500 consumers

1. BI builders collaborate in a controlled workspace. Consumers do not automatically need edit access to the workspace.
2. Create/use a governed semantic model with documented owner/freshness/security and reports built on it.
3. Choose app for broad polished distribution; use direct sharing for narrower cases when appropriate. Record licensing/capacity/tenant constraints before real rollout.
4. Promote/certify trusted content according to organizational governance, rather than letting five teams publish competing 'official revenue' models.
5. Define support owner, refresh owner, change notice and consumer path.

EXPECTED RESULT

An enterprise delivery plan separating building, governed semantic truth, distribution and support.

WHY IT WORKED

Workspace collaboration permissions are not confused with consumer distribution.

CLD5 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: Sales has 6 builders, 40 managers and 600 viewers. Design workspace + semantic model + app/share route.

COMMON BEGINNER MISTAKES

1. Give all consumers Contributor because app setup is unfamiliar. | 2. Every report owns its own copied semantic model. | 3. Certified means mathematically correct forever. | 4. No support/change owner after publish.

STUCK? RECOVERY

Draw Builders -> Workspace -> Semantic Model -> Report/App -> Consumers. Add owner/security/freshness at the model and support owner at distribution. Save the genuine attempt before review.

VALIDATE

Builders/consumers get minimum needed access; one authoritative KPI model is identifiable; distribution/support owners are explicit.

CLD5 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

Current PL-300 directly measures creating/configuring workspaces/apps, publishing/updating items, distribution methods, subscriptions/alerts and content endorsement. Tenant/license constraints must be checked at implementation time.

TEACH IT BACK

Explain why an app can be better than giving hundreds of people workspace access.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry introduces external guest/partner consumption and requires a policy check rather than assuming it works.

EVIDENCE / MASTERY

Builder-consumer architecture + distribution matrix + owner/freshness/security notes + explain-back.

VISUAL/SOURCE ROUTE: Optional PL-300 Manage and Secure video; current Microsoft collaboration docs control behavior.

https://learn.microsoft.com/en-us/power-bi/collaborate-share/collaborate-share-overview

CLD6 - Workspaces and deployment concepts

Page A - why, outcome, prerequisite, mental model and vocabulary

Developer -> DEV -> Validate -> TEST -> Approve -> PROD

WHY IT MATTERS

Editing production directly makes rollback, testing and impact analysis difficult. Enterprise BI separates development and released environments and controls who can promote changes.

WHAT YOU LEARN

Design DEV/TEST/PROD workspace separation, deployment ownership, pre-release validation, deployment-pipeline/CI-CD concepts and rollback/impact checks without pretending every tenant has the same tooling.

PREREQUISITE

CLD5 enterprise Service delivery and C2-04 lifecycle/change impact basics.

NEW WORDS BEFORE WE USE THEM

Environment: Separated development/testing/production context for items and configuration. Deployment pipeline: Fabric capability that helps manage content deployment across pipeline stages/workspaces. CI/CD: Practices/tools for controlled integration, testing and delivery of changes.

Promote a semantic-model/report change

1. Developer changes measure/report in DEV; production content is not edited directly.
2. Run model/reconciliation/report/security checks in DEV and then controlled TEST/user acceptance as required.
3. Review impact: downstream reports/apps/refresh/security/config. Record release approver.
4. Deploy/promote items using the organization's supported method, such as Fabric deployment pipelines and/or Git-integrated CI/CD where appropriate.
5. After release, verify PROD refresh/security/KPI smoke checks and define rollback/redeployment path. Pipeline permissions and workspace roles are separate concerns.

EXPECTED RESULT

A controlled release path with evidence before and after production promotion.

WHY IT WORKED

Changes are separated from production consumers until validation/approval is complete.

CLD6 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: change Customer Segment logic used by 12 reports. Build DEV/TEST/PROD route and an impact/release checklist.

COMMON BEGINNER MISTAKES

1. DEV and PROD are tabs in one workbook only. | 2. Deployment pipeline automatically tests business correctness. | 3. Contributor can deploy everywhere without separate pipeline/workspace permissions. | 4. No post-deploy smoke or rollback plan.

STUCK? RECOVERY

If tooling feels complex, design the manual governance first: separate environment -> validate -> approve -> deploy -> smoke -> rollback. Then map platform tools. Save the genuine attempt before review.

VALIDATE

No production edit occurs before validation; approver, dependencies, post-deploy smoke and rollback are explicit.

CLD6 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

Fabric deployment pipelines and broader CI/CD/Git capabilities evolve. Use current docs at implementation; preserve environment separation, review, validation and rollback as durable controls.

TEACH IT BACK

Explain deployment pipelines as release control, not a magical correctness checker.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry introduces a failed TEST validation and asks what must not reach PROD.

EVIDENCE / MASTERY

Environment map + release checklist + impact analysis + explain-back.

VISUAL/SOURCE ROUTE: Required current references: Fabric CI/CD overview and deployment-pipeline documentation.

https://learn.microsoft.com/en-us/fabric/cicd/cicd-overview

CLD7 - Refresh, gateways and scheduled operations

Page A - why, outcome, prerequisite, mental model and vocabulary

Source -> Credential -> Gateway? -> Refresh -> History -> Owner

WHY IT MATTERS

A report can be perfect at 9 PM and wrong at 9 AM if refresh credentials, gateway mapping or schedules fail. Freshness is an operational contract, not a checkbox.

WHAT YOU LEARN

Decide when an on-premises gateway is required; distinguish cloud/on-prem connections; design credentials, refresh cadence, failure ownership and freshness checks; avoid personal-account dependency.

PREREQUISITE

CLD2 identity, CLD5 Service and C2-04 lifecycle basics.

NEW WORDS BEFORE WE USE THEM

On-premises data gateway: Software bridge that lets cloud services securely reach supported data sources in a private/on-premises network. Scheduled refresh: Configured recurring refresh of a semantic model/data source connection. Freshness SLA: Maximum acceptable data age for the business use case.

CLD7 - Follow with me once

Page B - Enterprise SQL source to morning dashboard

Enterprise SQL source to morning dashboard

1. Classify data source location/connectivity. If Power BI/Fabric cannot reach the on-prem/private source directly, design a gateway/connection path.
2. Map all required source connections/credentials; one missing source mapping can make a gateway unavailable for the semantic model.
3. Choose refresh cadence from business need and source/load constraints—not 'as often as possible'.
4. Assign technical owner and business freshness owner; use non-personal/team-managed credentials where policy supports it.
5. After refresh, validate both technical success and business freshness/row/control totals. A successful refresh with stale source data is still a business failure.

EXPECTED RESULT

A refresh runbook with source connectivity, credential/gateway mapping, schedule, owners and post-refresh controls.

WHY IT WORKED

Network/credential/schedule/business freshness are tested separately.

CLD7 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: semantic model uses cloud SaaS + on-prem SQL + SharePoint. Decide gateway/credentials and create an 8 AM freshness runbook.

COMMON BEGINNER MISTAKES

1. Gateway required for every cloud source. | 2. Use analyst's personal password for production refresh. | 3. Refresh succeeded = source data current. | 4. No owner for expired credential/gateway outage.

STUCK? RECOVERY

Trace Source -> Network path -> Credential -> Gateway/connection -> Semantic model -> Schedule -> Freshness validation. Save the genuine attempt before review.

VALIDATE

Each source has correct connectivity method; owners and failure escalation exist; business freshness is checked after technical refresh.

CLD7 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

Current Power BI scheduled-refresh guidance separates gateway connection, credentials and schedule. Exact refresh limits/licensing vary; do not hard-code them into architecture without current verification.

TEACH IT BACK

Explain why a gateway is about connectivity to a source, not a speed booster.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry simulates expired SQL credential at 06:00 and requires recovery/escalation.

EVIDENCE / MASTERY

Connectivity map + refresh schedule/owner matrix + failure runbook + explain-back.

VISUAL/SOURCE ROUTE: Required visual: Power BI Gateway getting-started video; official gateway/refresh docs control current behavior.

https://learn.microsoft.com/en-us/power-bi/connect-data/refresh-scheduled-refresh

CLD8 - Permissions and governance

Page A - why, outcome, prerequisite, mental model and vocabulary

Identity -> Workspace role -> Item access -> Data security -> Catalog -> Policy

WHY IT MATTERS

Giving people access is not one checkbox. Workspace roles, item/semantic-model access, row-level security, OneLake/item security and governance/catalog signals solve different parts of the problem.

WHAT YOU LEARN

Apply least privilege across workspace/item/data layers; use groups; distinguish access control from RLS; add sensitivity/endorsement/catalog/governance responsibilities; create periodic access review.

PREREQUISITE

CLD2 identity, CLD5 Service and C2-04 RLS.

NEW WORDS BEFORE WE USE THEM

Workspace role: Admin, Member, Contributor or Viewer permission level within a Fabric workspace. Item-level access: Permission granted to a specific report, semantic model or Fabric item rather than whole workspace. Sensitivity label: Governance/compliance classification applied under organizational policy to help protect data/content.

Finance BI access design

1. Builders belong to appropriate security groups and get only required workspace roles.
2. Consumers receive app/item/semantic-model access rather than Contributor by convenience.
3. RLS limits rows inside the model for users who can query it; it does not replace workspace/item permission design.
4. Use OneLake catalog/endorsement/sensitivity/governance features to improve discovery, trust and policy visibility where the organization enables them.
5. Schedule access review: leavers, role changes, dormant broad access, external users and certified-content owners.

EXPECTED RESULT

Layered least-privilege design with access owner and recurring review.

WHY IT WORKED

Each security/governance control has a separate purpose and ownership.

CLD8 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: Finance builders, regional managers, executives and auditors need different actions/data. Build least-privilege matrix and quarterly review checklist.

COMMON BEGINNER MISTAKES

1. RLS means user can safely be workspace Admin. | 2. Viewer and app audience are identical concepts. | 3. Certified content means anyone may access it. | 4. Permissions assigned individually forever without group/review strategy.

STUCK? RECOVERY

Create columns: identity/group, resource, action, workspace role, item permission, data restriction, approval owner, review date. Save the genuine attempt before review.

VALIDATE

No persona gets broader access than needed; RLS and item access are not conflated; owner/review date exists.

CLD8 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

Current Fabric workspace roles are Admin/Member/Contributor/Viewer, while Fabric/Power BI also support item-specific and data-security mechanisms. Current PL-300 directly assesses workspace/item/semantic-model access, RLS and sensitivity labels.

TEACH IT BACK

Explain workspace permission versus RLS using 'can open the building' versus 'which records can you see once inside'.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry removes a regional manager and adds an external auditor under explicit policy review.

EVIDENCE / MASTERY

Least-privilege matrix + governance/catalog plan + access-review checklist + explain-back.

VISUAL/SOURCE ROUTE: Required current references: workspace roles, permission model, OneLake catalog Govern/Purview.

https://learn.microsoft.com/en-us/fabric/fundamentals/roles-workspaces

CLD9 - Monitoring and failure awareness

Page A - why, outcome, prerequisite, mental model and vocabulary

Job runs -> Monitor -> Detect -> Diagnose -> Recover -> Prevent repeat

WHY IT MATTERS

Enterprise analytics is not finished when the pipeline/report is published. Someone must detect failures, know the affected consumers, restore service and prevent recurrence.

WHAT YOU LEARN

Use run status/history and monitoring concepts to triage failed refresh/pipeline jobs; distinguish symptom/root cause; design incident evidence, severity, ownership, recovery and recurrence prevention.

PREREQUISITE

CLD7 refresh operations and C2-05 logging/validation concepts helpful but not required.

NEW WORDS BEFORE WE USE THEM

Monitoring: Observing operational health, status, history and signals from analytics jobs/items. Incident: Unplanned event that degrades correctness, freshness, access or availability. Root cause: Underlying condition whose correction prevents or reduces recurrence, not merely the visible symptom.

CLD9 - Follow with me once

Page B - Morning executive report is stale

Morning executive report is stale

1. Detect: report freshness check fails at 08:05. Record incident start, affected KPI/audience and required freshness SLA.
2. Check monitoring/refresh/job history: determine which job failed and capture exact status/error/time.
3. Trace dependencies: source availability -> credential/gateway -> ingestion/pipeline -> warehouse/model -> report. Find the first broken stage.
4. Recover safely: fix/renew/retry from appropriate step, then rerun downstream validation rather than only refreshing the final report.
5. Close incident with root cause, impact, evidence, owner and prevention action such as alert, credential rotation process, schema contract or capacity review.

EXPECTED RESULT

A repeatable incident runbook that connects monitoring signal to affected business output and safe recovery.

WHY IT WORKED

You diagnose dependencies from first broken layer instead of randomly clicking Refresh.

CLD9 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: nightly pipeline succeeds but semantic-model refresh fails; build incident timeline, root-cause tree and recovery/prevention actions.

COMMON BEGINNER MISTAKES

1. Retry until green without root cause. | 2. Only record screenshot of final report. | 3. No business-impact/severity field. | 4. Fix pipeline but never revalidate downstream KPI/security/freshness.

STUCK? RECOVERY

Use five boxes: Impact -> First failed job -> Dependency cause -> Recovery -> Prevention. Do not skip impact/validation. Save the genuine attempt before review.

VALIDATE

Incident record includes status/error/time, affected consumers/KPI, first broken dependency, safe rerun path and prevention owner.

CLD9 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

Fabric's Monitor/monitoring hub currently centralizes job execution health/history. Teams may add alerts/SLAs/ticketing outside Fabric; the course assesses the operational thinking, not one UI.

TEACH IT BACK

Explain why 'Refresh failed' is a symptom, not automatically the root cause.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry uses a pipeline that succeeds technically but produces stale source data.

EVIDENCE / MASTERY

Incident report + dependency tree + recovery validation + prevention action + explain-back.

VISUAL/SOURCE ROUTE: Required current reference: Fabric monitoring hub; optional Learn Live Data Factory pipelines video.

https://learn.microsoft.com/en-us/fabric/admin/monitoring-hub

CLD10 - Analyst - analytics engineer - data engineer collaboration

Page A - why, outcome, prerequisite, mental model and vocabulary

Business question -> Analyst -> Analytics eng -> Data engineer -> Contract -> Shared evidence

WHY IT MATTERS

A senior analyst should not throw vague tickets at engineers or silently rebuild engineering logic in Power BI. Strong teams agree on ownership, grains, freshness, definitions and failure evidence.

WHAT YOU LEARN

Distinguish typical responsibilities without rigid job-title stereotypes; write usable data contracts/issues/change requests; identify when the analyst should self-serve, escalate or collaborate.

PREREQUISITE

All C2-07 plus C2-06 lineage/quality and C2-05 reproducibility concepts.

NEW WORDS BEFORE WE USE THEM

Data contract: Explicit agreement about dataset structure, grain, keys, freshness, quality expectations and ownership. Handoff: Transfer of a requirement/change/incident with enough evidence for the receiving role to act. Escalation: Raising a problem to the role/team that owns the affected system or decision.

CLD10 - Follow with me once

Page B - Revenue dashboard breaks after upstream schema change

Revenue dashboard breaks after upstream schema change

1. Analyst proves impact: Revenue KPI changed after 07:00 refresh; records expected/observed values and affected report/consumers.
2. Trace lineage to staging table where RegionCode renamed/changed. Do not immediately hard-code a workaround in the report.
3. Data/analytics engineering owners receive a reproducible issue: source/object, schema diff, grain/key, first failing transformation, sample IDs, freshness/SLA and desired contract.
4. Engineering repairs or intentionally changes upstream contract; analyst validates business semantics and downstream KPI after redeploy.
5. Team records ownership/prevention: schema-change notification, tests, versioned contract or consumer impact process.

EXPECTED RESULT

A handoff that another technical role can act on without a meeting just to discover basic facts.

WHY IT WORKED

Business impact and technical evidence meet at a shared contract rather than being thrown over a team boundary.

CLD10 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: Support SLA dashboard needs a new event field unavailable downstream. Decide which role owns definition, ingestion, transformation, semantic exposure and business validation.

COMMON BEGINNER MISTAKES

1. Analyst = dashboards only, engineer = everything technical. | 2. Opening 'data is wrong' ticket with no expected/observed evidence. | 3. Rebuilding official transformations locally to avoid collaboration. | 4. Engineer owns KPI business definition by default.

STUCK? RECOVERY

Use issue template: Business impact -> expected/observed -> dataset/item -> grain/key -> lineage point -> freshness -> evidence -> requested action -> owner/deadline. Save the genuine attempt before review.

VALIDATE

Every deliverable/decision has one accountable owner; handoff includes reproducible evidence and agreed acceptance check.

CLD10 - Prove mastery

Page D - professional version, teach-back, protected review and evidence

THEN SHOW THE PROFESSIONAL VERSION

Current PL-300 audience guidance explicitly expects Power BI analysts to collaborate with analytics engineers and data engineers to identify/acquire data. Exact job boundaries vary by company; contract/evidence quality is the portable skill.

TEACH IT BACK

Explain analyst, analytics engineer and data engineer collaboration as overlapping responsibilities connected by contracts—not three isolated ladders.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry is a schema-deprecation notice requiring impact analysis and coordinated migration.

EVIDENCE / MASTERY

RACI/ownership map + issue/change template + acceptance validation + explain-back.

VISUAL/SOURCE ROUTE: Book-led; current PL-300 audience profile and Fabric collaboration context support the lesson.

https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/pl-300

C2-07 mastery checklist + quick reference

Use after the first pass; live UI exposure never replaces reasoning

TEN QUESTIONS

CLD1 Can you separate tenant/workspace/capacity/items? | CLD2 Is failure compute, identity, credential or data permission? | CLD3 Why does each Fabric workload exist? | CLD4 Who owns authoritative OneLake items and why copy/shortcut? | CLD5 How do builders and consumers differ? | CLD6 What prevents an unvalidated change reaching PROD? | CLD7 Does every source have connectivity/credential/refresh/freshness ownership? | CLD8 Which workspace/item/data/governance control grants or restricts access? | CLD9 What is the first broken dependency and recovery/prevention? | CLD10 Can another role act on your handoff without rediscovering the problem?

BLOCK MASTERY IF

Admin everywhere; tenant/workspace/capacity confused; every Fabric workload added; uncontrolled data copies; consumers given build roles; direct PROD edits; personal refresh credentials; gateway misunderstood; RLS treated as all security; incident only retried; vague 'data wrong' engineering ticket.

FAST OPERATIONAL DEBUG

Identity -> resource/action -> workspace/item/data permission -> source connectivity/credential -> job/refresh -> monitoring history -> business freshness/output -> owner/incident/prevention. Find first broken layer.

OPTIONAL PATH

C2-07 is optional Senior/BI enterprise depth. It overlaps heavily with later engineering operations, but direct Course 1 -> C2B -> C3 learners receive required cloud/engineering prerequisites in their own route.

C2-07 source / video registry

Current Microsoft docs control changing platform behavior

CLD1-4 - What is Microsoft Fabric?
https://learn.microsoft.com/en-us/fabric/get-started/microsoft-fabric-overview
Current SaaS/workloads/OneLake overview; updated 2026-03-31

CLD4 - What is OneLake?
https://learn.microsoft.com/en-us/fabric/onelake/onelake-overview
Tenant-wide logical lake; current 2026 reference

CLD3-4 - Microsoft OneLake documentation
https://learn.microsoft.com/en-us/fabric/onelake/
OneLake security/catalog/shortcuts navigation

CLD5 - Collaborate and share Power BI content
https://learn.microsoft.com/en-us/power-bi/collaborate-share/collaborate-share-overview
Workspaces/apps/share/endorsement

CLD5-6 - Workspaces in Power BI
https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-new-workspaces
Current enterprise workspace reference

CLD6,8 - Roles in Fabric workspaces
https://learn.microsoft.com/en-us/fabric/fundamentals/roles-workspaces
Admin / Member / Contributor / Viewer

CLD6 - Fabric CI/CD overview
https://learn.microsoft.com/en-us/fabric/cicd/cicd-overview
Current release/source-control concepts

CLD6 - Deployment pipelines
https://learn.microsoft.com/en-us/fabric/cicd/deployment-pipelines/intro-to-deployment-pipelines
Managed stage deployment concepts

CLD7 - Configure scheduled refresh
https://learn.microsoft.com/en-us/power-bi/connect-data/refresh-scheduled-refresh
Gateway connection, credentials and schedules

CLD7 - On-premises data gateway
https://learn.microsoft.com/en-us/data-integration/gateway/service-gateway-onprem
Current gateway concept

CLD7 - Gateway visual
https://www.youtube.com/watch?v=Vlv1tbuZNcM
Optional visual; official docs control current behavior

CLD8 - Fabric permission model
https://learn.microsoft.com/en-us/fabric/security/permission-model
Fabric access/security model

CLD8 - Govern Fabric data with OneLake catalog
https://learn.microsoft.com/en-us/fabric/governance/onelake-catalog-govern
Governance posture/recommended actions

CLD8 - Microsoft Purview + Fabric
https://learn.microsoft.com/en-us/fabric/governance/microsoft-purview-fabric
Governance/security/compliance integration

CLD9 - Fabric monitoring hub
https://learn.microsoft.com/en-us/fabric/admin/monitoring-hub
Centralized job health/history

CLD5-8,10 - PL-300 study guide
https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/pl-300
Skills measured as of Apr 20 2026

CLD5-6 - PL-300 Manage and Secure video
https://learn.microsoft.com/en-us/shows/exam-readiness-zone/preparing-for-pl-300-manage-and-secure-power-bi
Optional exam-alignment visual

CLD3-5 - Power BI Meets Microsoft Fabric
https://learn.microsoft.com/en-us/shows/fabric-tech-talk-fridays/power-bi-meets-microsoft-fabric-whats-in-it-for-you
Optional platform visual

VERIFY / FALLBACK

Rechecked 2026-09-10. External media is support only; internal lessons and synthetic fixtures remain sufficient if tenant/video access fails.